Ethanol Blending in Gasoline – Belgium

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Ethanol Blending in Europe

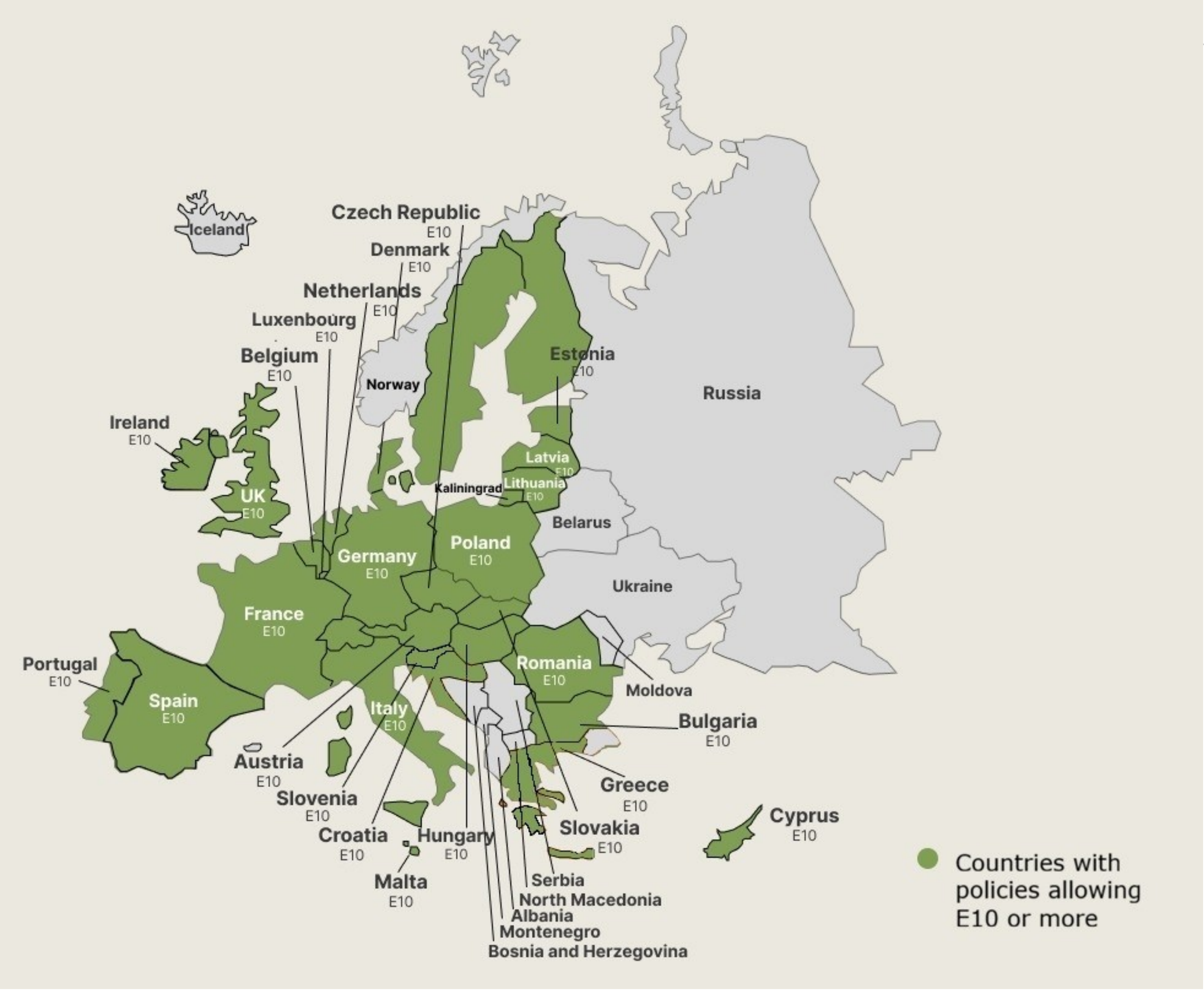
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.

- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.

- Ethanol contributes to transport decarbonization and air quality improvement.

- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

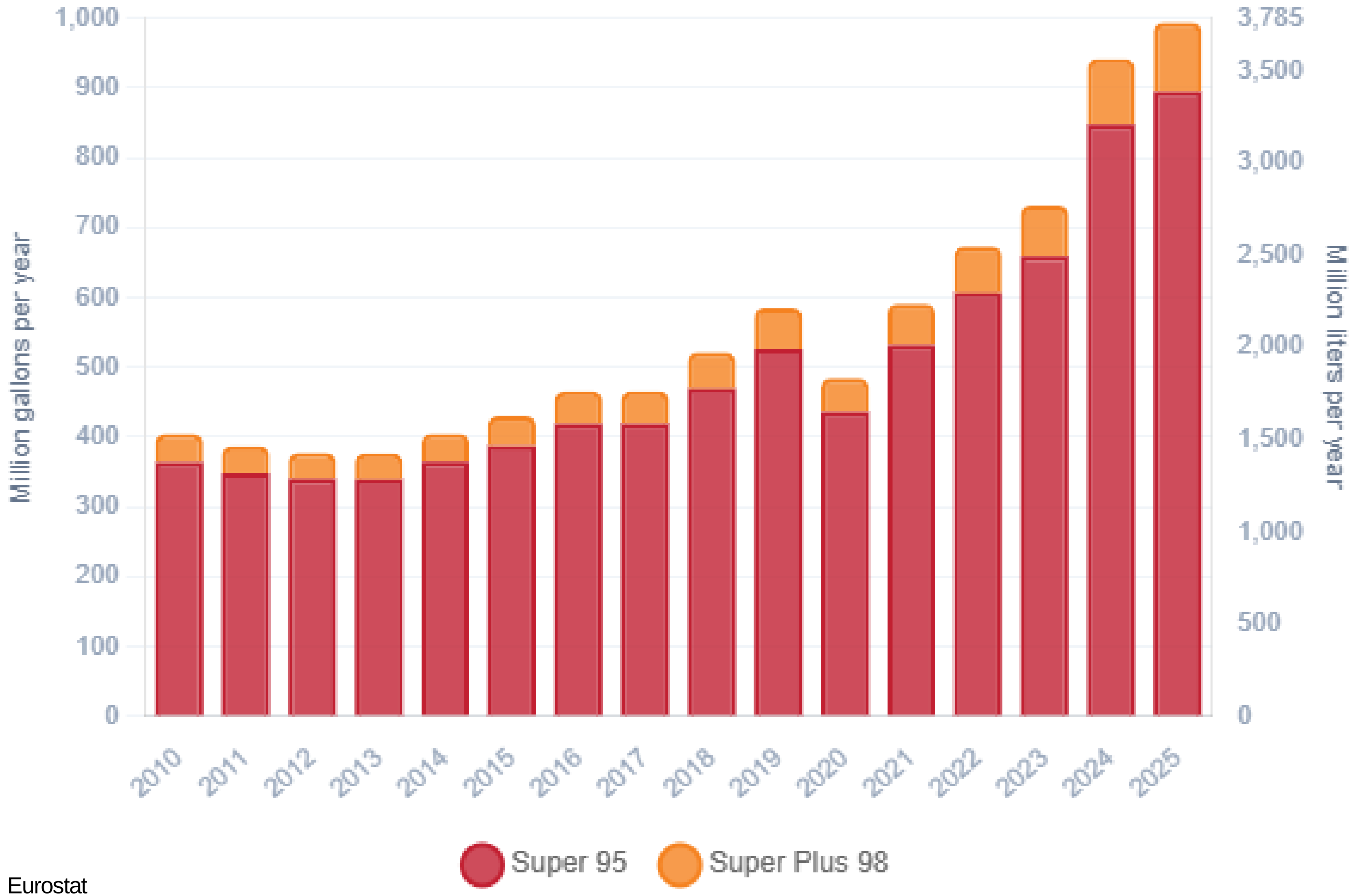
Ethanol Gasoline Blending in Belgium

In 2025, gasoline consumption in Belgium was 3,761 million liters (993 million gallons) and growth rate was 6.2%. Gasoline production and net imports were 4,397 and -4,172 million liters (1,162 and -1,102 million gallons) correspondingly. Belgium complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2025, ethanol consumption in Belgium was 165 million liters (43 million gallons) representing 4% of gasoline demand. Ethanol production and Net Imports were 620 and -105 million liters (164 and -28 million gallons) correspondingly. Ethanol sources are Sugar Cane 20% and Cereals 80%.

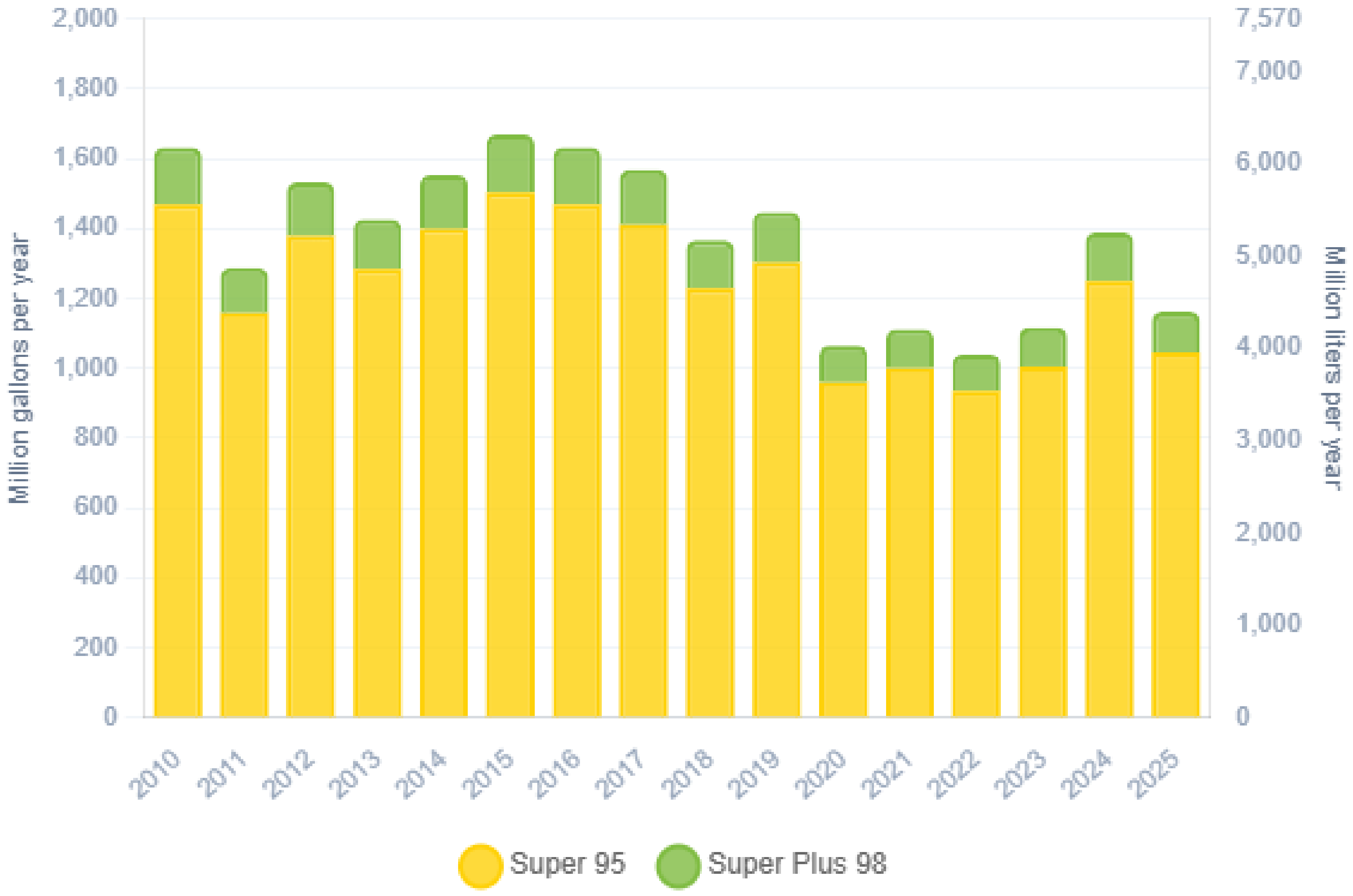
Source: Eurostat, European Commision

Gasoline Demand in Belgium



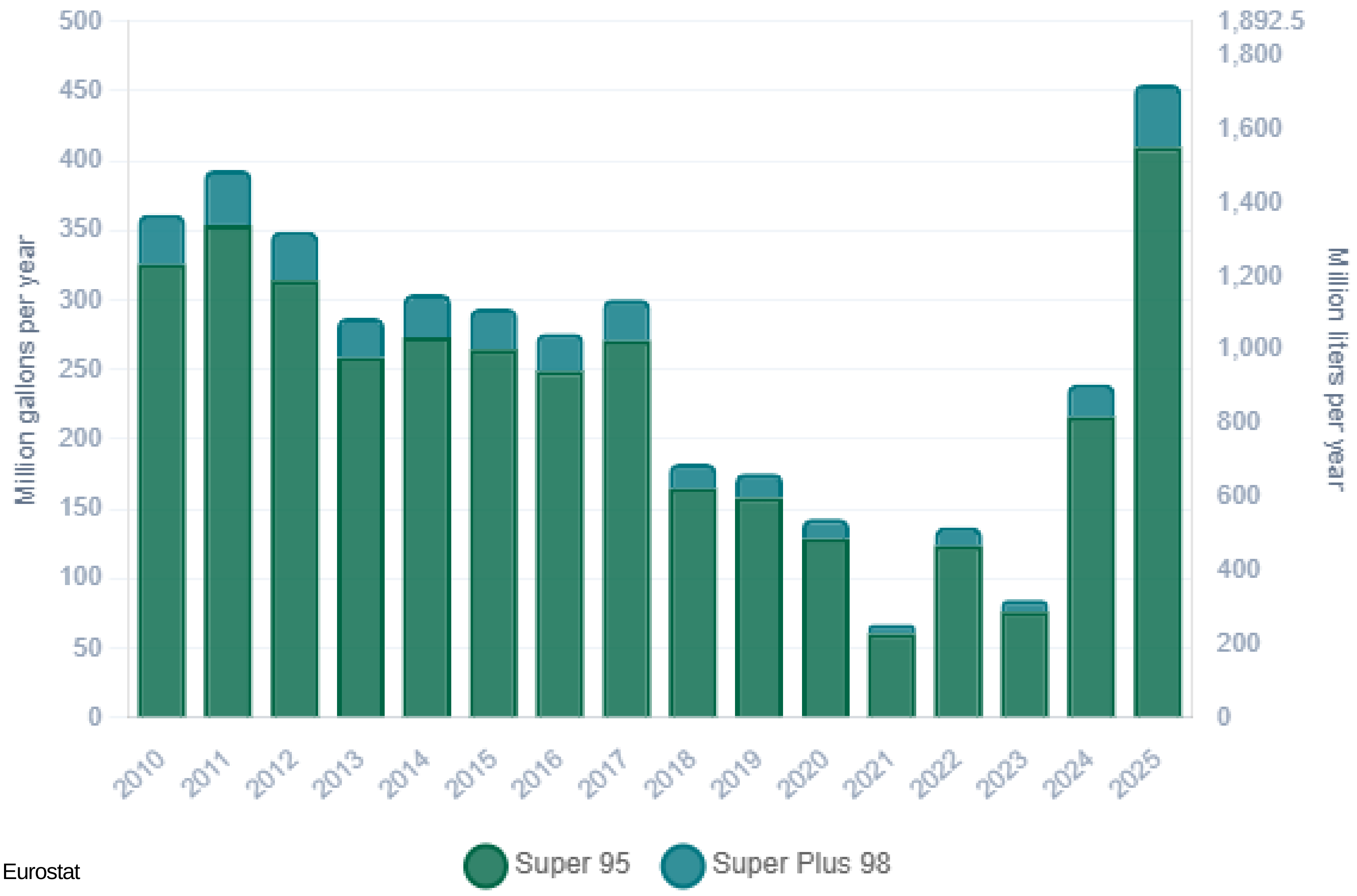
Source: Eurostat

Gasoline Production in Belgium



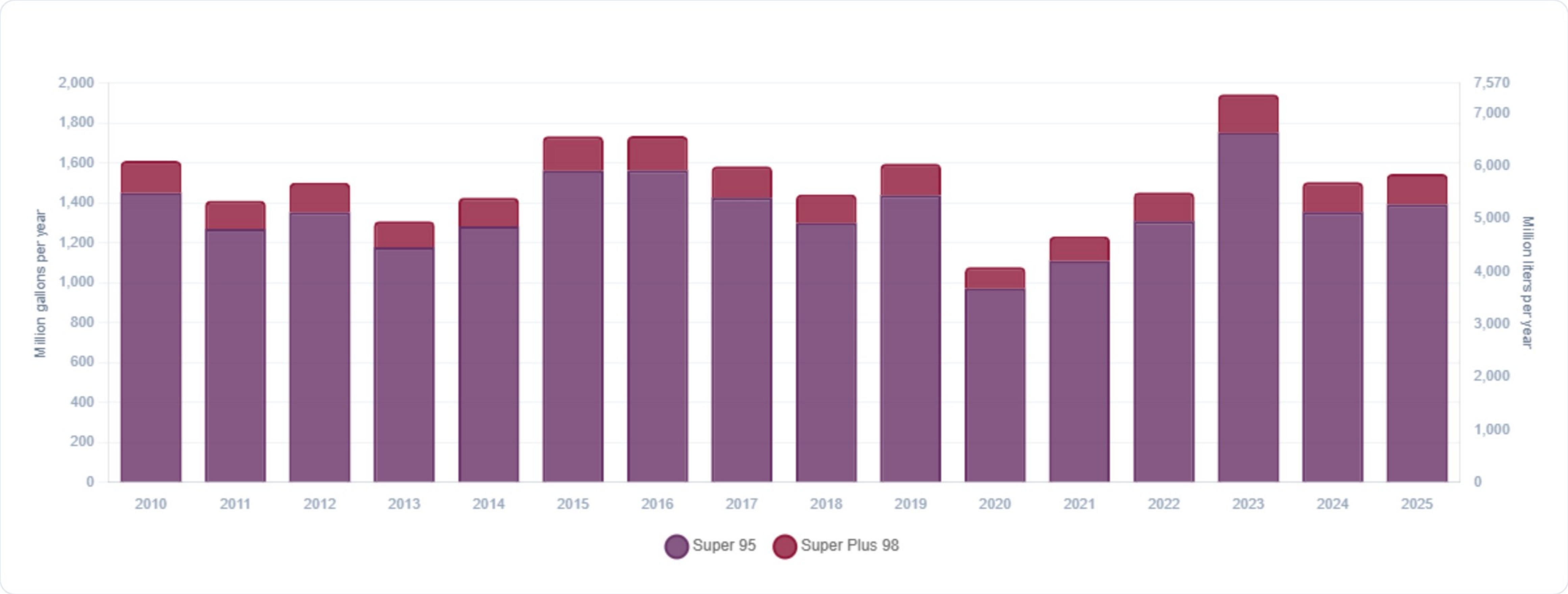
Source: Eurostat

Gasoline Imports in Belgium



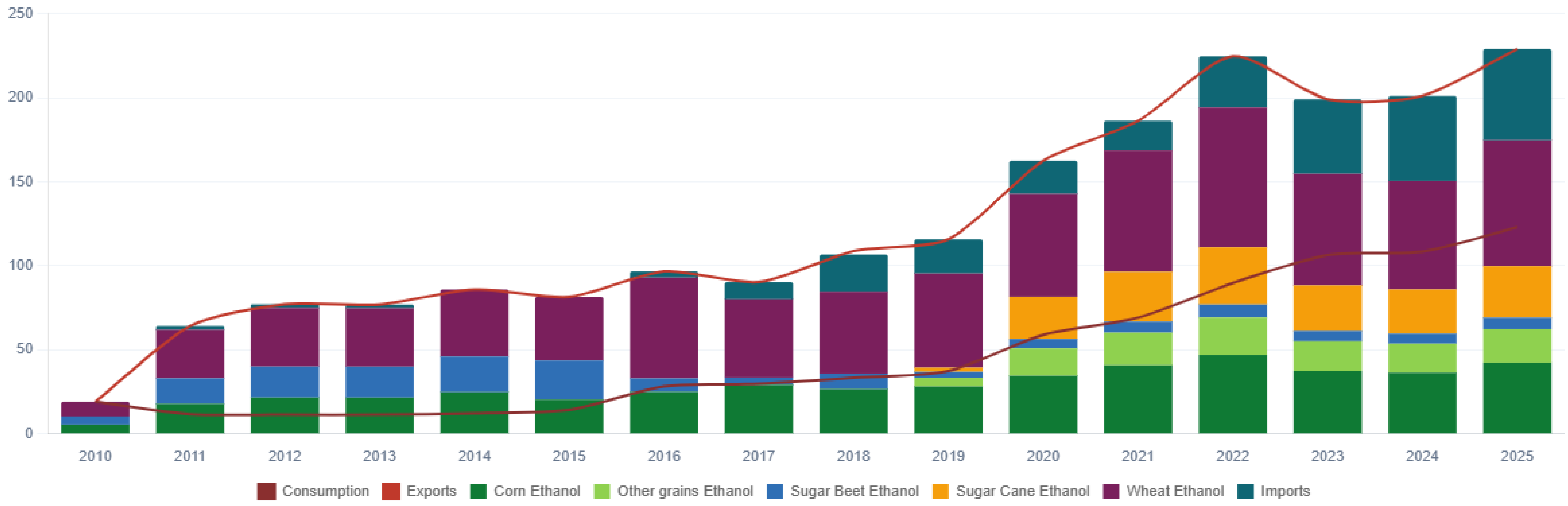
Source: Eurostat

Gasoline Exports in Belgium



Source: Eurostat

Ethanol Balance in Belgium



Source: Eurostat

Gasoline Quality in Belgium

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.5							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	5.7							
GHG Emission Reduction (%)	-							

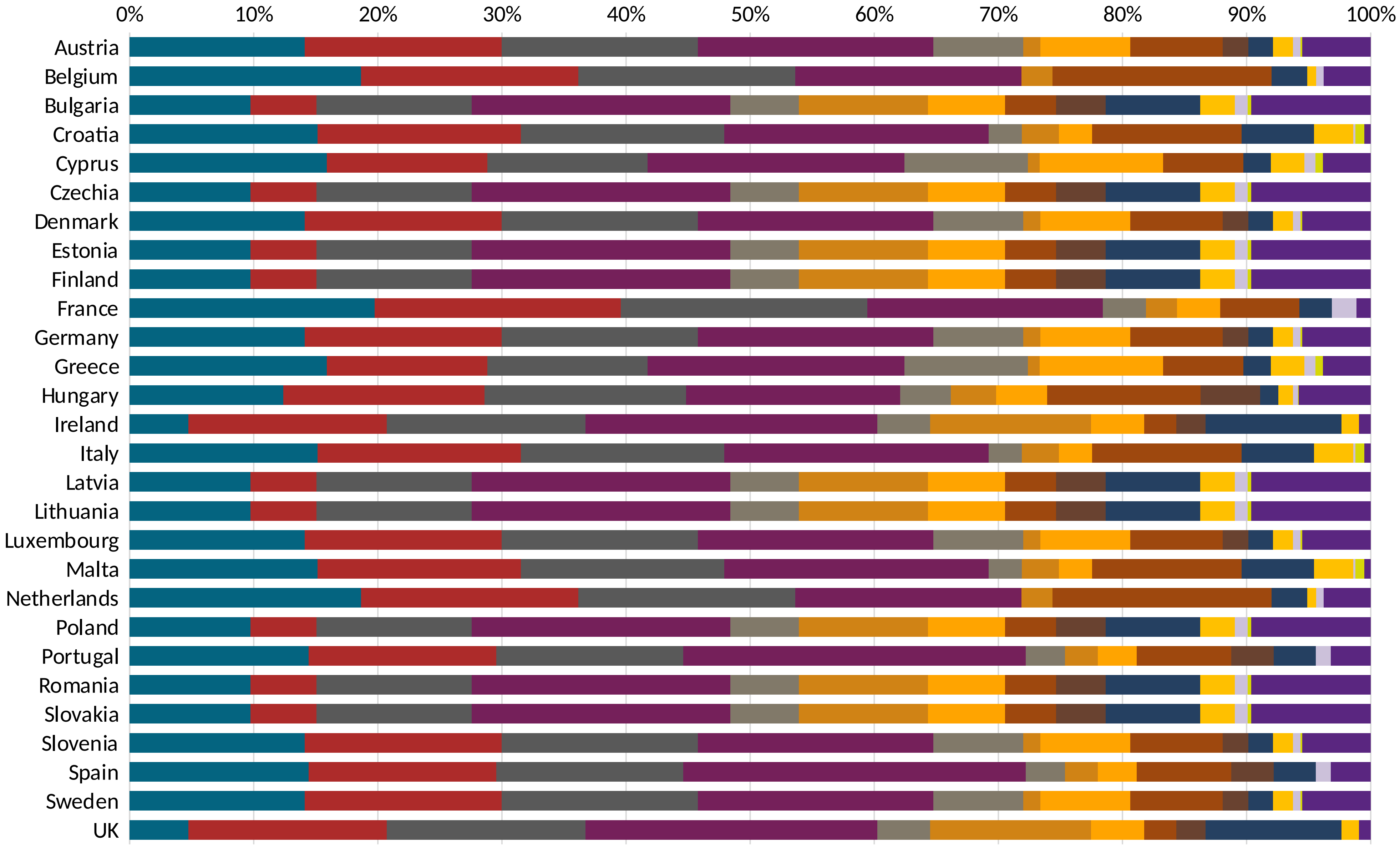
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

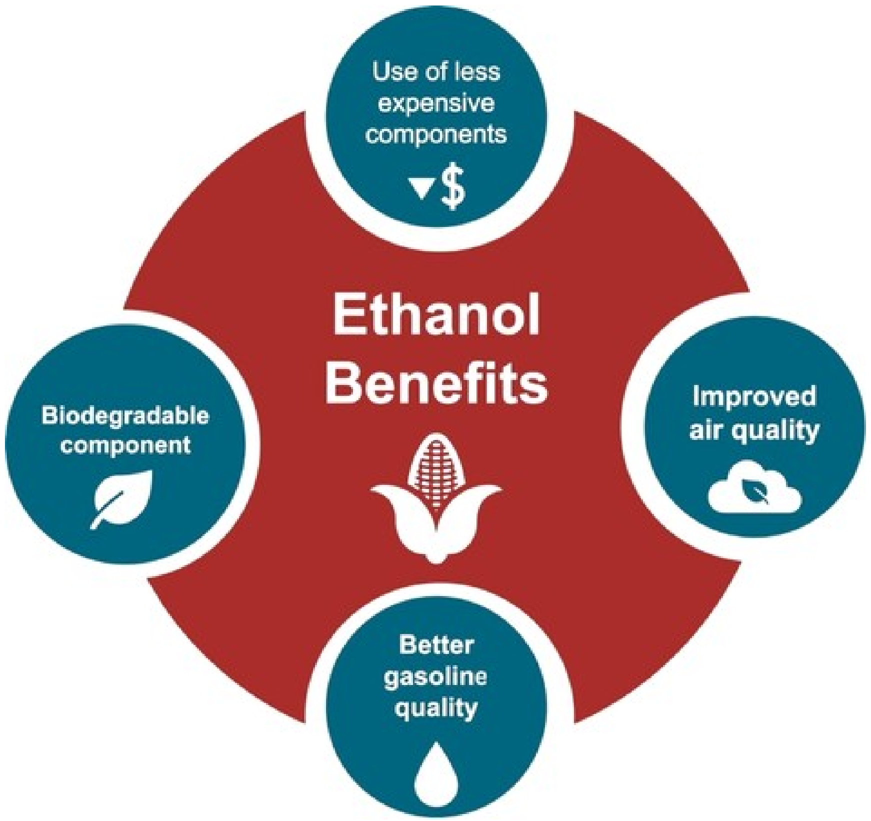
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

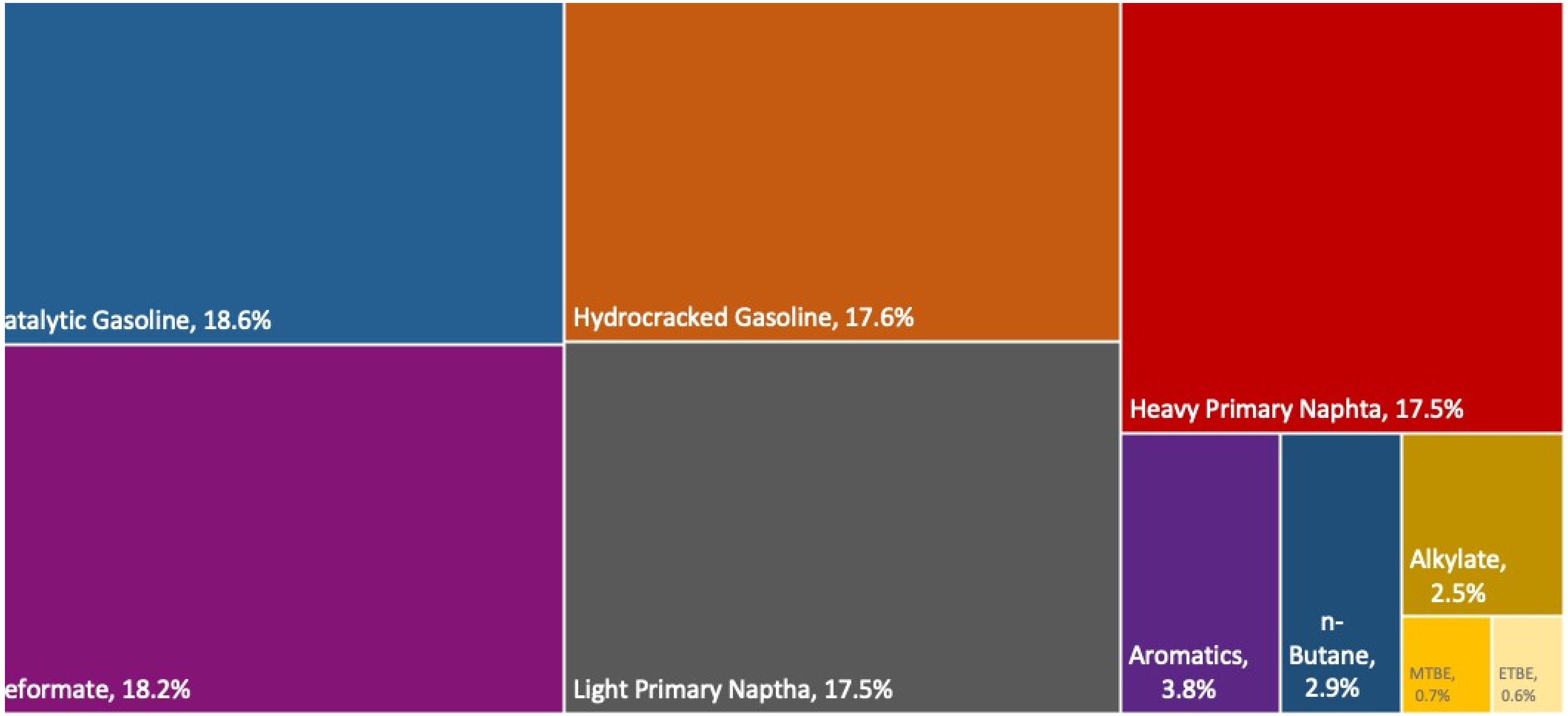
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

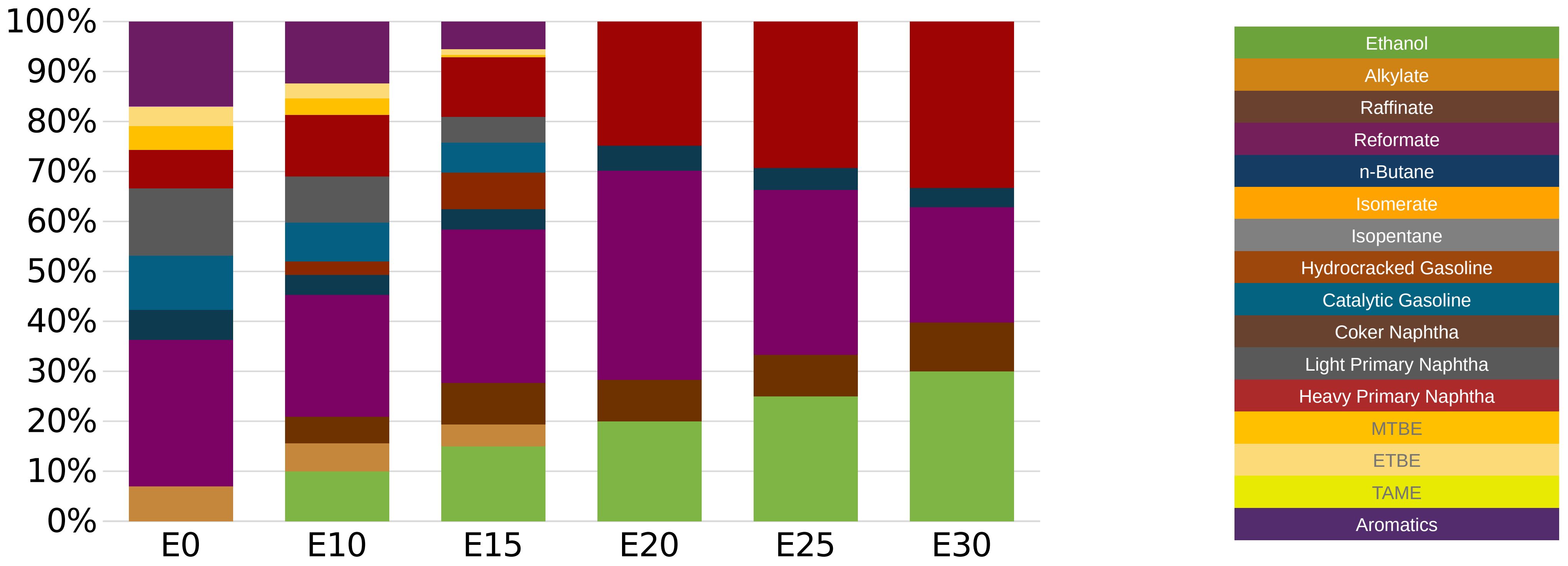


Belgium Available Blending Components



Source: Faro90

Belgium – Gasoline 95 – Constant Octane

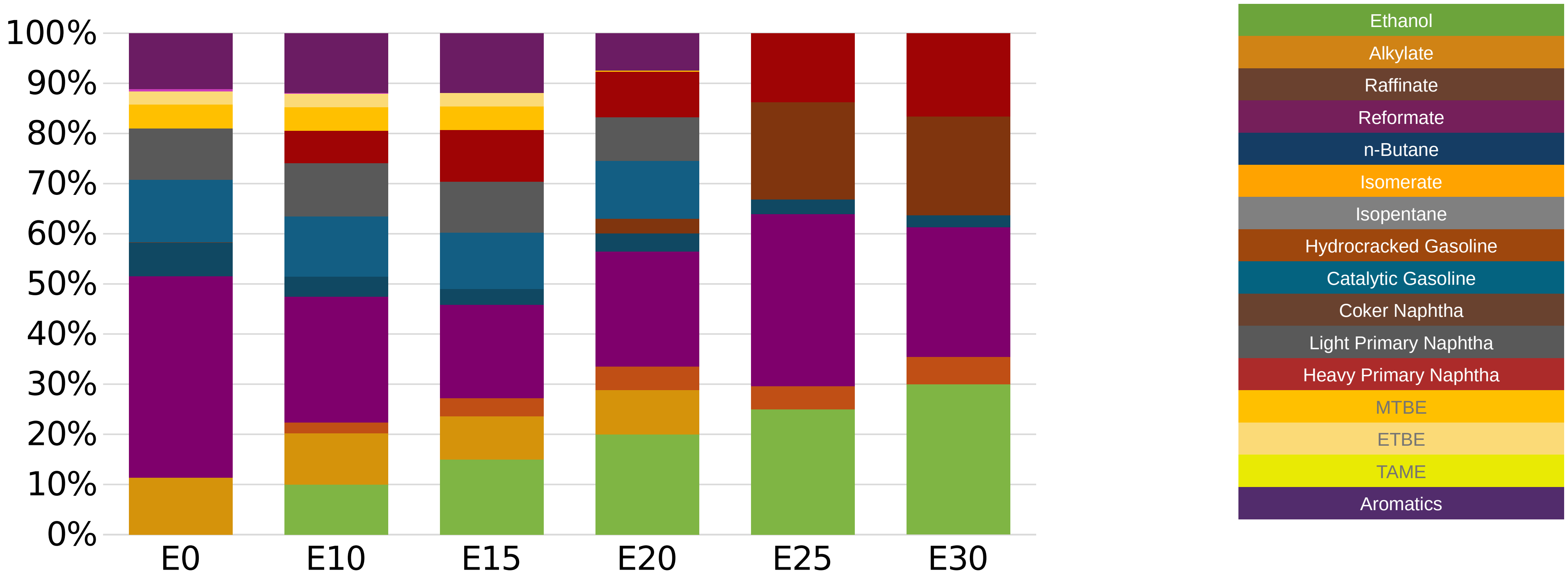


Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US$/gal)	$ 2.078	$ 1.944	$ 1.869	$ 1.748	$ 1.672	$ 1.577

Average CIF 2025 Prices

Source: Faro90

Belgium - Gasoline 98 - Constant Octane

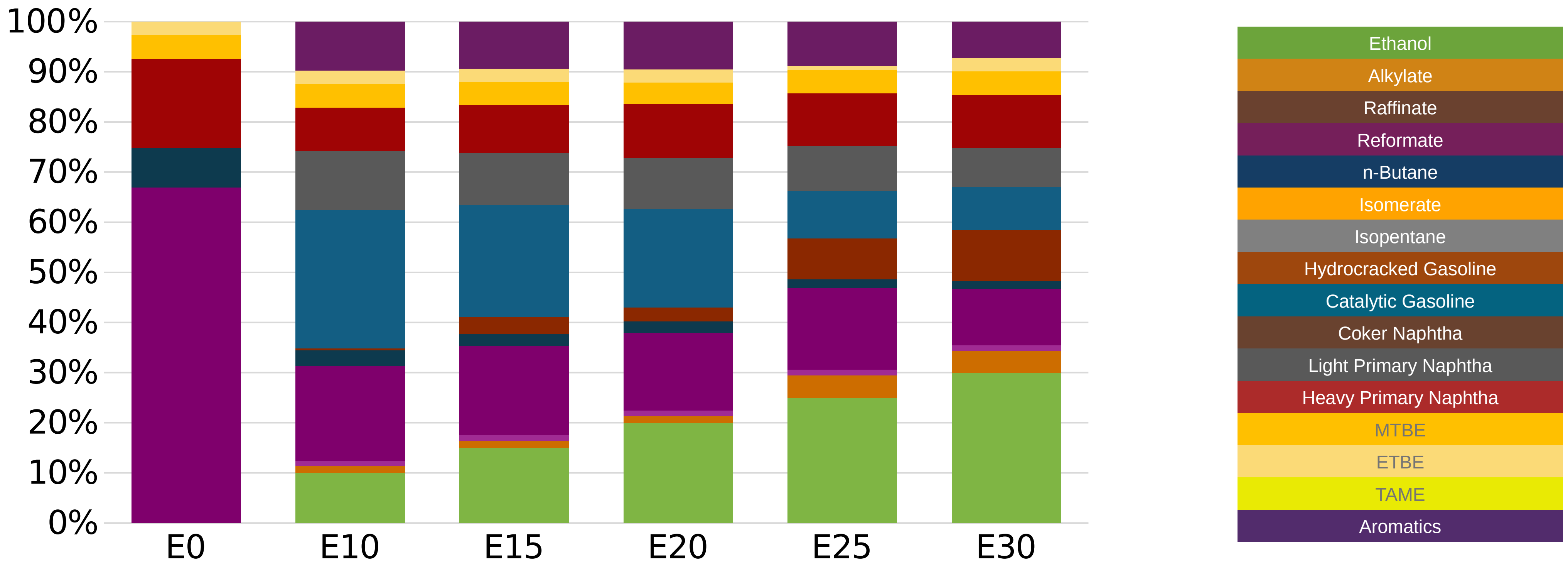


Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US$/gal)	$ 2.206	$ 2.087	$ 2.013	$ 1.961	$ 1.854	$ 1.791

Average CIF 2025 Prices

Source: Faro90

Belgium – Gasoline 95 – Increased Octane

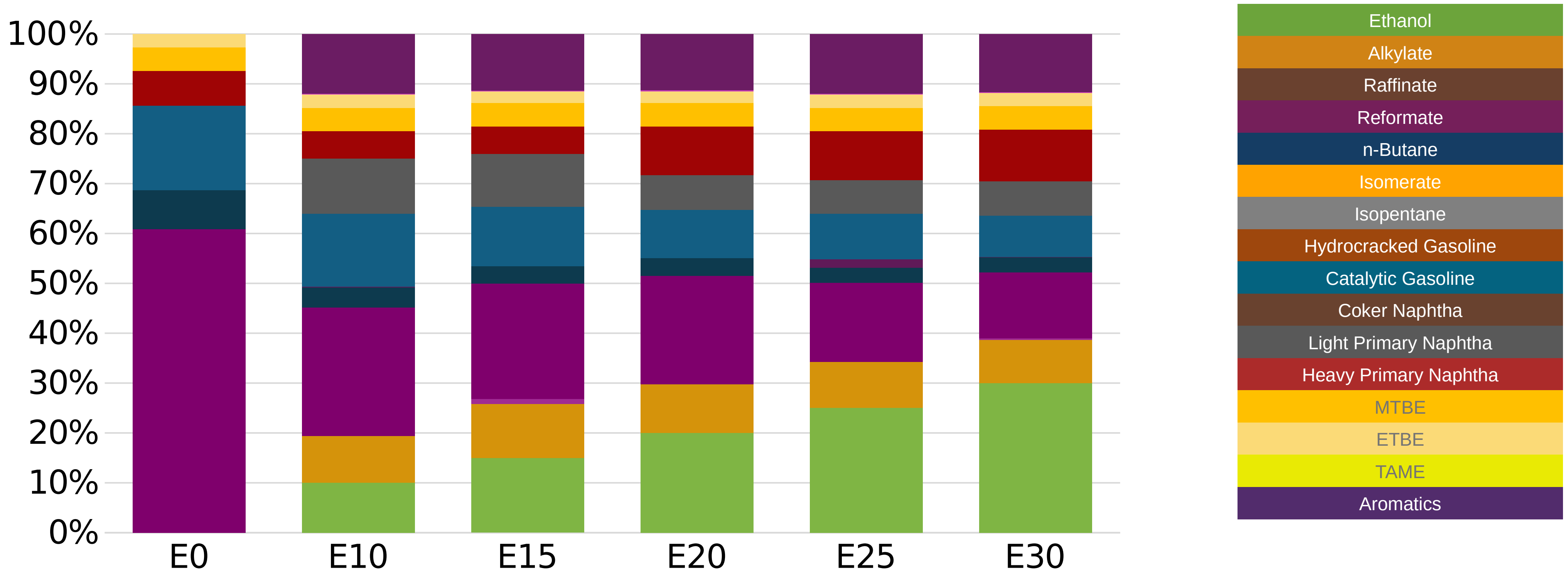


Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US$/gal)	$ 1.990	$ 1.990	$ 1.990	$ 1.990	$ 1.990	$ 1.990

Average CIF 2025 Prices

Source: Faro90

Belgium – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US$/gal)	$ 2.126	$ 2.126	$ 2.126	$ 2.126	$ 2.126	$ 2.126

Average CIF 2025 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**

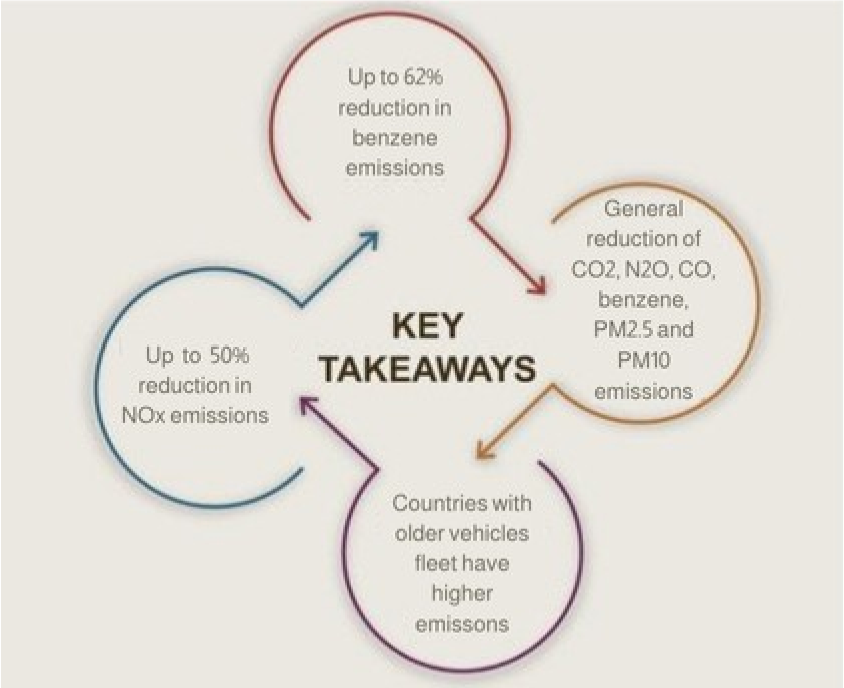
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet - Belgium

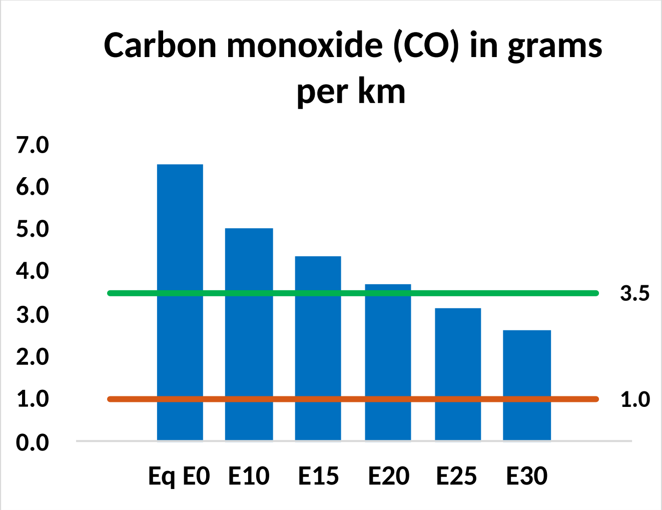
Type	Weight	Engine	Exhaust	Recovery	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	7,069
		Single FI	3-Ways		>8 years	40,824
		Multi FI	3-Ways		>8 years	838,076
			Euro III		>8 years	351,981
			Euro IV		>8 years	918,689
			Euro V		>8 years	18,921
			Euro 6		>8 years	676,845
					4-8 years	382,031
					<4 years	181,023
			Hybrid		All	760,491
		Electric			All	296,606
	Medium	Carburator	3-Ways	PCV Tank	>8 years	923
		Single FI	3-Ways		>8 years	1,212
		Multi FI	3-Ways		>8 years	104,128
			Euro III		>8 years	101,978
			Euro IV		>8 years	413,943
			Euro V		>8 years	2,696
			Euro 6		>8 years	297,812
					4-8 years	229,449
					<4 years	182,967
			Hybrid		All	229,446
		Electric			All	107,135
Sml Engines	Light	4-Cycle	None	PCV	>8 years	4,936
			Euro III		>8 years	8,395
			Euro IV		>8 years	32,397
			Euro V		>8 years	16,529
			Euro 6		>8 years	90,103
					4-8 years	22,658
					<4 years	375,620
			Hybrid		All	4
		Electric			All	4,414
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	42,584
		Fuel-Injection	3-Ways		>8 years	481,263
			Euro III		>8 years	269,910
			Euro IV		>8 years	311,133
			Euro V		>8 years	87,098
			Euro 6		>8 years	120,243
					4-8 years	56,690
					<4 years	7,365
			Hybrid		All	9,882
		Electric			All	15,685

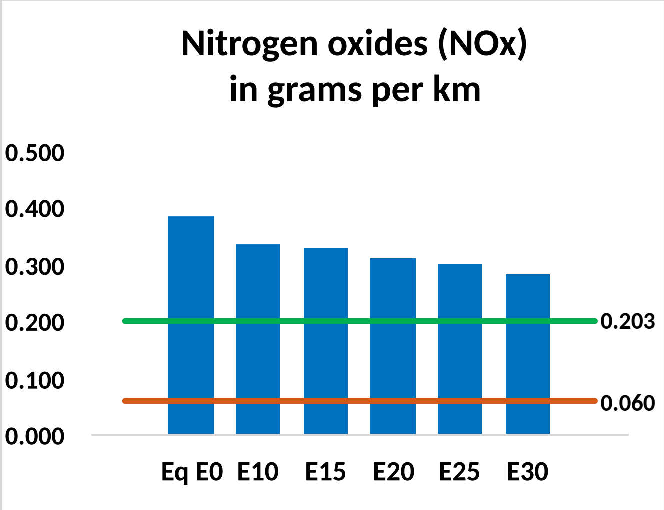
Vehicle Fleet: 8,101,154
Gasoline Vehicle Fleet: 3,299,143

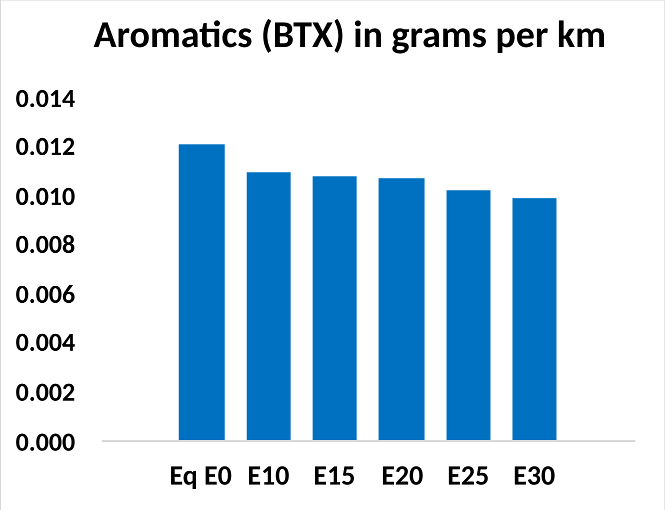
Source: Eurostat, ACEA

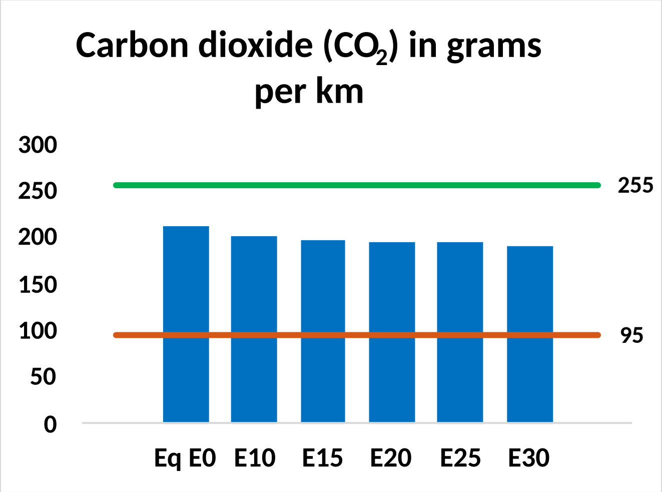
Belgium – Vehicular Emissions

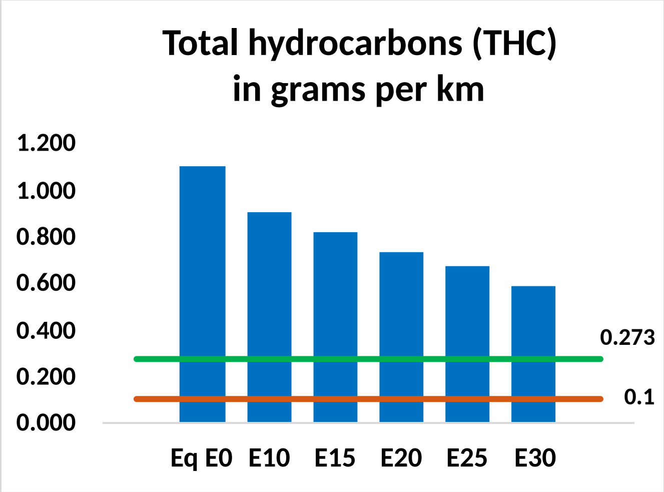
TIER III BIN 125 United States
Euro 6 Standard

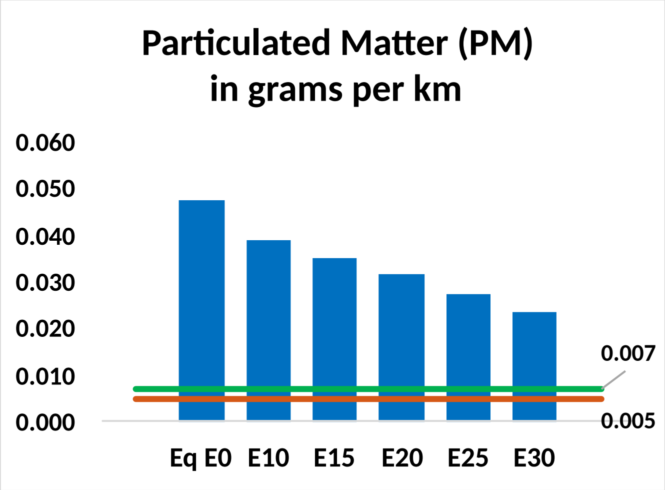












Source: Faro90

Belgium – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	554,479	490,937	427,395	371,243	312,858	267,397	221,422	-23%	-44%
CO2	17,996,550	17,533,782	17,096,723	16,752,989	16,582,895	16,482,982	16,179,153	-5%	-8%
VOC	93,593	85,189	76,784	69,661	62,384	56,755	49,360	-18%	-33%
NOx	33,022	30,490	28,823	28,009	26,658	25,618	24,381	-13%	-19%
SOx	205	189	174	161	148	135	120	-15%	-28%
NH3	6,828	6,760	6,693	6,725	6,800	6,853	6,898	-2%	0%
N2O	353	351	349	351	358	362	366	-1%	2%
CH4	18,449	18,449	18,449	18,726	19,132	19,482	19,814	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	480	451	422	401	379	363	341	-12%	-21%
Acetaldehyde	1,511	1,735	2,545	3,876	5,281	6,034	7,130	68%	249%
Formaldehyde	5,654	5,497	6,408	7,524	7,883	8,720	9,493	13%	39%
Benzene	1,026	986	934	919	912	872	844	-9%	-11%
PM2.5	1,036	942	848	765	684	597	506	-18%	-34%
PM10	3,022	2,748	2,473	2,231	1,994	1,741	1,475	-18%	-34%

Source: Faro90

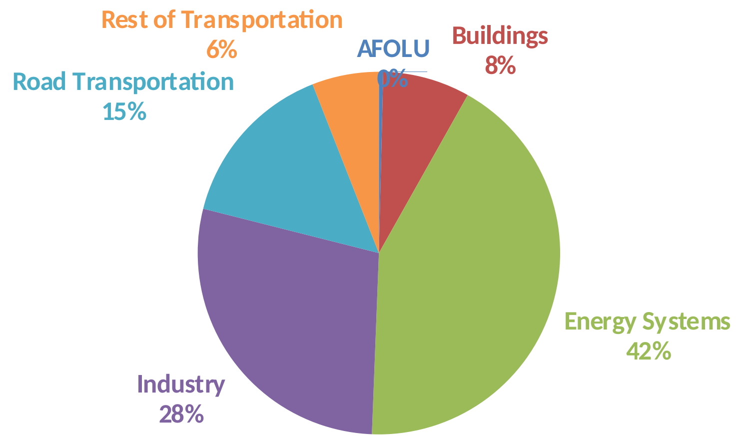
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

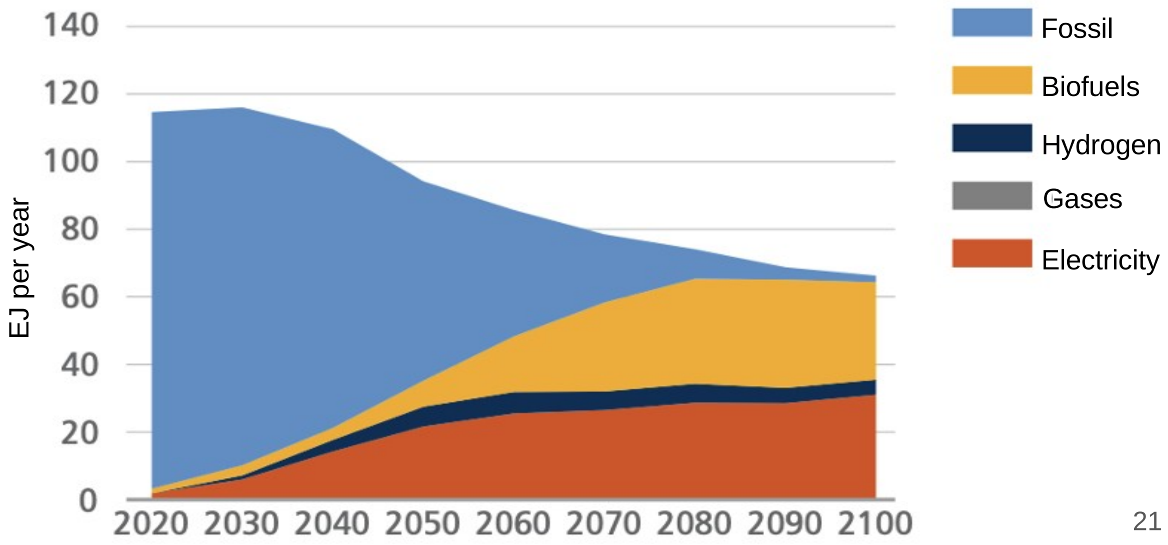
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

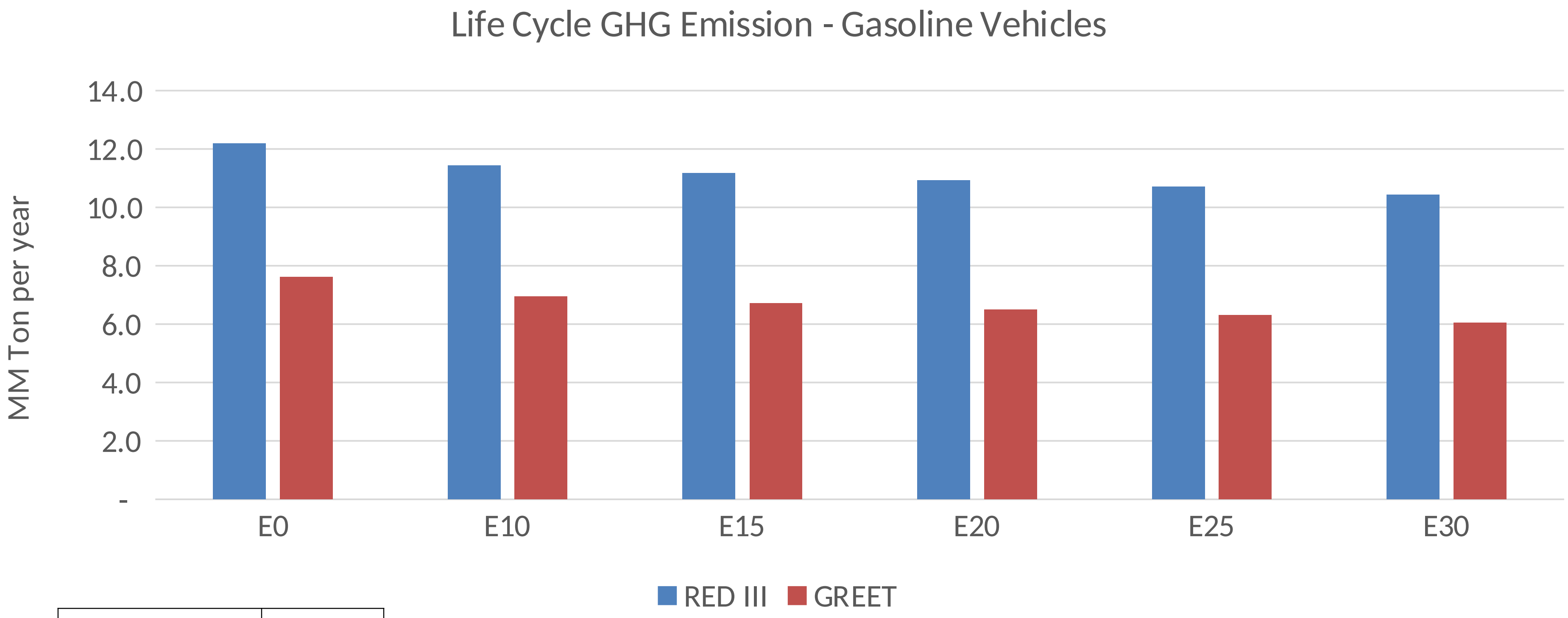
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Belgium – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	105.6
Target @ 2035 (MMT/year)	59.4
Delta	46.1

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90